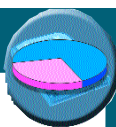


## How Much Space is Enough?



Depending on your computing needs, users can be classified into the following types:

### Less is good...

...if your time on the PC is spent surfing the Internet and maybe create a spreadsheet or two. A hard disk with the bare minimum capacity should be enough for you. The same holds true for businesses where the PC is primarily used for accounting and billing purposes, or for creating letters and spreadsheets. A 20 GB drive should suffice your needs, though 20 GB drives are fast phasing out. Preferably, go for a 40 GB hard disk—this is fast emerging as an entry-level solution. Given your frugal need for space, a 40 GB hard disk will last you several years.

### Size does matter...

...if you're a typical home user who stores the latest applications, MP3s, videos, movie trailers and even full-length movies, not to mention space-hogging games. No amount of space is ever going to satisfy you. Go for at least 60 GB of space. If you feel you might exceed that as well, then 80 GB might be more appropriate. Space also matters if you use CAD/CAM software, or are a graphics professional with a penchant for 3-D animations.

### Speed is your mantra...

...if you are a power user, but lack the purchasing power to buy a SCSI drive. Consider the 7200-rpm drives—the capacity you can get would depend on how much you can afford, but an 80 GB drive should suit you just fine.

drives from Seagate came with a leather cover to protect the drive electronics and the sensitive internal chips. The 7200-rpm drives from Seagate too came with a screwed on back cover to protect the drive electronics. Seagate calls this feature Seashell and it is of great help in protecting the drives during the installation and transportation process, especially from static discharge. There were no manuals accompanying the Seagate drives.

The Samsung SpinPoint drives use proprietary features such as NoiseGaurd and claims to run at 3-decibel noise levels. Indeed, these drives were exceptionally quiet during the tests—a certain advantage if you work at night or in an environment where any small noise can be disturbing.

As far as design goes—a factor that many users are not too bothered with—all Samsung drives had the same design. The only confusing part with Samsung drives was the jumper setting—two jumpers are required for setting the drive as master and some users might find this confusing.

These general features aside, we also evaluated the drive specifications and bundled accessories.

**Up to 60 GB capacity:** Capacity is one of the major parameters on which a hard disk is bought today. The decision to buy a higher capacity drive is based solely on what you are going to use it for. Judging by the current trend, 20 GB drives will soon be phased out, as bigger drives get more affordable. Even if you don't require too much disk space, it would be wiser to go in for at least a 40 GB drive—it costs just

marginally more than a 20 GB drive. A 40 GB drive is a good starting point as it provides sufficient space for holding games, general applications and even media files for home and non-demanding office uses. But if you store huge amounts of data, you need at least 60 GB.

Surprisingly, we received just one 60 GB drive in this comparison—the 5,400-rpm Samsung SV0602H. Other than this there were three 20 GB drives and four 40 GB drives. The Samsung SV0602H is priced just Rs 300 more than the SV4002H, the 40 GB drive of the same spindle speed from Samsung, though it was in the middle of the pack in terms of performance.

The 20 GB drive from Maxtor, the 541DX, was the sleekest drive in this comparison, but it was also the noisiest.

If you are looking for something smaller and lighter on the pocket, then the Samsung SV2001H is a good deal. This 20 GB drive is available for less than Rs 3,800, though the catch is that its performance is not quite up to the mark.

The Samsung SP2001H, however, hits the sweet spot in this category—this 7,200-rpm drive offers 20 GB space but showed an impressive performance, and will please those in search of performance and good price rather than capacity.

All the drives featured a 2 MB data buffer and they all sported a single-platter design with an ATA/100 interface. None of the drives in this category included accessories.

**Over 60 GB capacity:** For gamers and MP3 addicts, a drive that has less than 60 GB storage could prove to be restrictive, especially if you are forever downloading songs and movies—a single DivX rip can take up 700 MB.

Since most games released today consume at least half a GB of hard disk space, you need as much space as you can get if you want the 'I-have-the-most-number-of-games-installed' boasting rights. And, of course, if you are into graphics or video editing, you know you need as large a hard disk as possible.

We received 10 drives that offered storage capacity in excess of 60 GB. Here, too, all the drives featured a 2 MB data buffer. However, there was a wide variation in the number of platters used to implement these drives—they ranged from the two-platter design of the IBM 82.3 GB drive to a four-platter design sported by the Maxtor 160 GB drive. Interestingly, all the Western Digital drives were based on an unusual three-platter design.

The Maxtor drives come with mounting plates and screws. Also provided is Maxtor's proprietary disk management software called Max-Blast, which allows you to manage your drives, partitions and prepare them for loading the operating system during installation.

The Samsung and Maxtor drives also feature installation documentation, unlike any of the other drives in this category.

At 200 GB, the WD2000 from Western Digital was the largest drive. Another mammoth was the 160 GB drive from Maxtor. This was also the only drive that featured the new ATA/133 drive interface—all the others in this comparison were based on

the ATA/100 interface

The 200 GB Western Digital WD2000 delivered excellent performance and is a great buy for the user who demands mind-blowing performance coupled with oodles of storage space. However, this drive tended to run really hot—it would be a good

**info** The total amount of storage space of the hard disks we tested in this comparison amounted to 1.3423 Terabytes—enough to store about 312 full-length DVD movies with Dolby Digital sound!